Comparative metrics and ablation study

All numeric results below are computed from saved observations. No additional model inference or human annotations were invented. CSV and JSON companions support reproduction and later analysis.

Metrics and availability

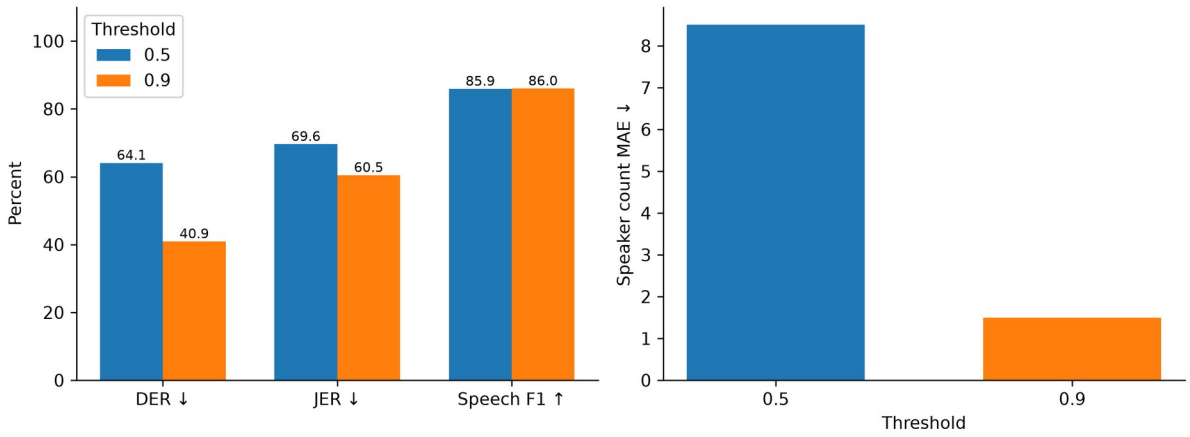
Component	Metrics	Current evidence
Speech recognition	WER, CER; substitutions/deletions/insertions; paired error differences	Measured English AMI and Hindi/Hinglish public pilots
Speaker diarization	DER, missed/false-alarm/confusion components, JER	Measured on two held-out AMI excerpts
Speech activity	Union-of-speech precision, recall and F1	Measured from the same diarization output; ignores identity
Speaker count	Absolute count error and meeting-averaged MAE	Measured on the same two excerpts
Slide alignment	Precision, recall, F1	Implemented; independent reference slide IDs needed
Event candidates	Precision, recall, F1 by event type	Implemented; independent reference events needed
Decisions and lineage	Decision P/R/F1; node/edge P/R/F1	Implemented; independent references needed
Influence ranking	Spearman and Kendall against human ratings	Implemented; independent ratings needed
Efficiency	Stage runtime, RTF, process RSS	Existing runs measured; cache/concurrency scopes must be respected
Functional correctness	API, browser, Python and Android checks	Functional tests, separate from model accuracy

Measured audio comparison

Threshold	DER %	JER %	Speech P %	Speech R %	Speech F1 %	Speaker count MAE
0.5	64.08	69.60	81.68	90.57	85.89	8.50
0.9	40.91	60.45	81.70	90.73	85.98	1.50

DER weights speaker-time, while JER averages reference-speaker errors; report both. Speech detection collapses identities to speech regions, so high speech F1 does not establish correct speakers. Count MAE also cannot establish correct attribution. The same 0.25-second collar and 120-second UEM are used. These are additional exploratory diagnostics; the original threshold was selected using development DER only. Definitions follow the pyannote.metrics documentation.

The new comparison is visualised below.



Model comparison

Corpus	Model	Ref words	S	D	I	WER %	CER %	Relative WER reduction %
hindi	small	208	88	6	12	50.96	20.09	0.00
hindi	medium	208	72	2	3	37.02	14.98	27.36
hinglish	small	124	60	4	18	66.13	50.00	0.00
hinglish	medium	124	49	8	14	57.26	52.25	13.41

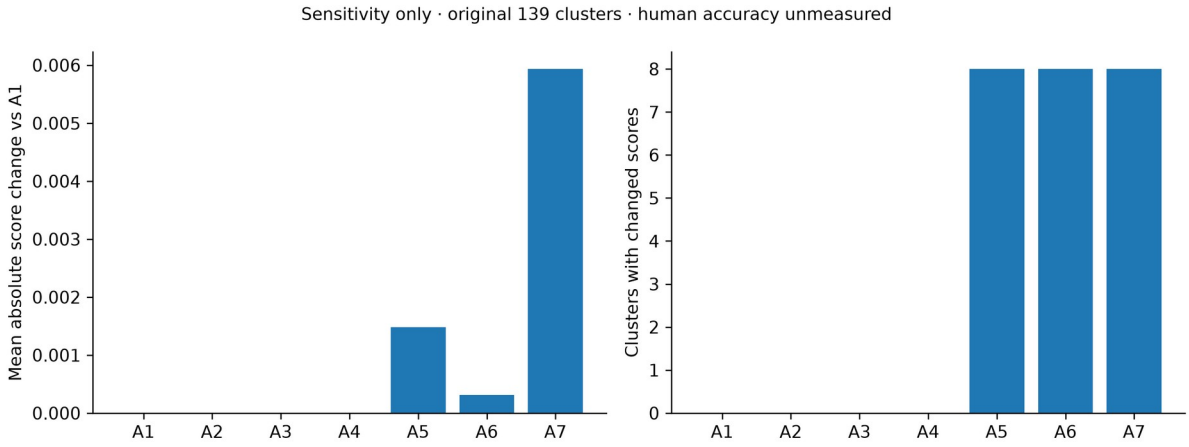
Each corpus comparison uses the same eight samples and normalisation. Do not pool Hindi and Hinglish into an overall product accuracy. Medium improves WER here but increases Hinglish CER. This is a model comparison; changing model size is not a component-removal ablation.

Component ablation and sensitivity

ID	Variant	Slide-grounded events	Changed scores	Mean absolute score change
A1	full_system	6	0	0.000000
A2	without_ppt	0	0	0.000000
A3	without_temporal_alignment	6	0	0.000000
A4	without_semantic_alignment	0	0	0.000000
A5	without_evidence_features	6	8	0.001484
A6	without_interaction_features	6	8	0.000312
A7	without_decision_impact_features	6	8	0.005935
A8	speaking_time_only_baseline	N/A	N/A	N/A

A1 is the control. A2 removes presentation links and reruns downstream NLP. A3/A4 rerun alignment and downstream analysis after removing temporal/semantic evidence. A5–A7 remove feature weights, then renormalize the remaining weights. A8 replaces the scoring method with speaking time. Counts and score movement are descriptive. Human accuracy remains unmeasured in every row.

The chart below excludes A8 because its scoring definition differs. Mean change averages all 139 clusters, including zeros, which can make a change appear small. Consult changed-score counts and maximum change in the CSV. Fragmentation in this original report means these diagnostics must not support claims about real participants.



A subsequent feature audit found that only decision contribution is active: proposed scores equal 0.15 times the decision-count baseline. A5/A6 changes are entirely weight renormalisation and preserve rankings; none of the 34 slides has recorded timing. Read

RESEARCH_VALIDITY_REVIEW.md before interpreting these charts. A zero change is a legitimate result. It can indicate absent features or no usable temporal information, rather than successful ablation. Threshold tuning is a calibration study; A2–A7 are the component-removal study. Neither is evidence of superior influence accuracy without independent labels.

Baseline comparison

Method	Clusters	Nonzero scores	Human ranking accuracy
B1_speaking_time	139	139	Unmeasured
B2_speaking_time_turns	139	139	Unmeasured
B3_decision_event_count	139	8	Unmeasured
B5_transcript_only	139	8	Unmeasured
B4_decision_evidence_count	139	0	Unmeasured
B6_audio_ppt_no_decision	139	139	Unmeasured
Proposed_decision_linked	139	8	Unmeasured

Six baseline implementations and the proposed method are available. Nonzero-score coverage describes outputs, not correctness. Baseline score scales differ, so score magnitudes are not compared as accuracy. No comparison with commercial systems or published paper scores is made unless the same audio, references and protocol are actually run.

Reproduce

From the project root:

```
.venv/bin/python scripts/build_comparative_evaluation.py
.venv/bin/python scripts/build_research_documents.py
.venv/bin/python scripts/check_research_package.py
```

The first command re-scores saved AMI predictions and reads completed downstream variants; it does not rerun neural inference. To recompute downstream variants for an independently annotated meeting:

```
.venv/bin/python scripts/run_evaluation.py --meeting-id YOUR_MEETING_ID --annotations
YOUR_ANNOTATIONS.json --ablation --output-dir outputs/evaluation/YOUR_RUN
```

Human raters must annotate before inspecting model outputs. Keep speaker/session splits and report rater agreement. Missing reference fields remain null, and graphs must not plot

them as zero.